

Use brackets to change the order

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $240 \div (8 + 4)$. Copy or complete the first step.

2. Calculate $(75 - 27) \times 6$.

3. Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.

4. Miro says: Miro ignores brackets because division normally comes first. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $240 \div (8 + 4)$. Copy or complete the first step.
Answer 20.
2. Calculate $(75 - 27) \times 6$.
Answer 288.
3. Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.
Answer 56 and 120.
4. Miro says: Miro ignores brackets because division normally comes first. Is this correct? Explain.
Answer Brackets override the usual priority and must be completed first.

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Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $240 \div (8 + 4)$.

6. Check this result using an inverse, estimate or known fact: Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.

2. Calculate $(75 - 27) \times 6$.

7. Place one pair of brackets in $100 - 20 \div 4 + 1$ to make the answer 21.

3. Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.

8. Identify and correct this misconception: Miro ignores brackets because division normally comes first.

4. Solve this independently and show every stage: Calculate $240 \div (8 + 4)$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $(75 - 27) \times 6$.

10. Write and solve a similar question to: Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.

Teacher answers and checking prompts

1. Calculate $240 \div (8 + 4)$.
Answer 20.
2. Calculate $(75 - 27) \times 6$.
Answer 288.
3. Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.
Answer 56 and 120.
4. Solve this independently and show every stage: Calculate $240 \div (8 + 4)$.
Answer 20.
5. Use a second method or representation for: Calculate $(75 - 27) \times 6$.
Answer Expected result: 288. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.
Answer Expected result: 56 and 120. A valid check must be shown.
7. Place one pair of brackets in $100 - 20 \div 4 + 1$ to make the answer 21.
Answer $(100 - 20) \div 4 + 1 = 21$.
8. Identify and correct this misconception: Miro ignores brackets because division normally comes first.
Answer Brackets override the usual priority and must be completed first.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Place one pair of brackets in $100 - 20 \div 4 + 1$ to make the answer 21.

6. Create a new example that tests the same idea as: Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.

2. Prove the answer to this question, rather than only calculating it: Calculate $240 \div (8 + 4)$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $(75 - 27) \times 6$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 56 and 120.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro ignores brackets because division normally comes first.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Place one pair of brackets in $100 - 20 \div 4 + 1$ to make the answer 21.
Answer $(100 - 20) \div 4 + 1 = 21$.
2. Prove the answer to this question, rather than only calculating it: Calculate $240 \div (8 + 4)$.
Answer Expected result: 20. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $(75 - 27) \times 6$.
Answer Expected result: 288. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 56 and 120.
Answer One valid reconstruction is: Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro ignores brackets because division normally comes first.
Answer Brackets override the usual priority and must be completed first.
6. Create a new example that tests the same idea as: Compare $16 + 8 \times 5$ with $(16 + 8) \times 5$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.